INCH-POUND

MIL-P-63142B (AR)

31 October 1990

SUPERSEDING
MIL-P-63142A (AR)
8 February 1981

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for loading, assembling and packing for the 8 inch, HE, RA, M650 projectile including the motor body and warhead (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1320

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection By Attributes
MIL-sTD-129	- Marking and Shipment for Storage
MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification
MIL-STD-453	- Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.1 Other Government documents, drawings, and publicaitons. The following other Government documents, drawings, and publicaitons form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.12)

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9287992	- Rocket Motor, 8 Inch, M650
9287993	- Warhead, 8 Inch, HE, M650
9287994	- Projectile, HE, RA, 8 Inch, M650
9288021	- Booster Auxiliary PA-500 & Forward Warhead Section Assembly
9288022	- Aft Warhead Section, Loading Assembly
9280132	- Pallet for Projectile, HE, RA, M650

(Copies of drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting officer.)

Code of Federal Regulations
Title 49 - Transportation parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scraps from floor sweepings ventilation and suction apparatus (other than from drilling operations specifially controlled to prevent contamination of explosive) shall not be used (see 6.5).

3.2 Assembled projectile. The assembled projectile shall comply with all requirements specified on Drawing (dwg.) 9287994, and all secondary drawings and with all requirements specified in the applicable specificaitons.

3.2.1 Motor body. No foreign matter shall be within the interior of the motor body prior to loading of propellant grains.

3.2.2 Delay. There shall be no lubricant on the face of the delay assembly or rocket cap assembly.

3.2.3 Propellant grain. The surface of the propellant grain shall be free of chips, cracks, gouges, dirt, grease or other foreign material.

3.2.4 Rotating band. The surface of the rotating band lands shall be examined for surface damage.

a. On the forward two lands, there shall be no evidence of raised copper due to nicks, digs or damage of any kind.

b. On the aft three lands, any nicks or digs extending 1/2 the land width or less are acceptable, provided the maximum band diameter is not exceeded.

3.3 Cavitation in high explosive charge.

3.3.1 Cracks and projected cavities. Cracks and cavities within the explosive charge shall not exceed the requirements speicified in Table I when determined in accordance with 4.4.2.9. Cavities having a maximum (max) projected length of 1/32 inch (in) or less shall be disregarded.

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TABLE I. Cavitations in high explosive

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks, and annular rings, square inch (sq. in.)	*1/64	1/4	1/2	1/2
Projected length of any cavity, excluding pipes, cracks, and annular rings (in.)	1/8	1/2	1/2	3/4
Piping cavities, max. projected area (sq. in.)	Ø	1/4	1/2	---
Piping cavities, max. projected width (in.)	Ø	1/4	1/4	---
Cracks, max. projected width (in.)	1/32	1/32	1/16	---
Annular rings, max. projected width (in.)	Ø	Ø	1/4	---

*If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch. Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times greater than the width.

An Annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

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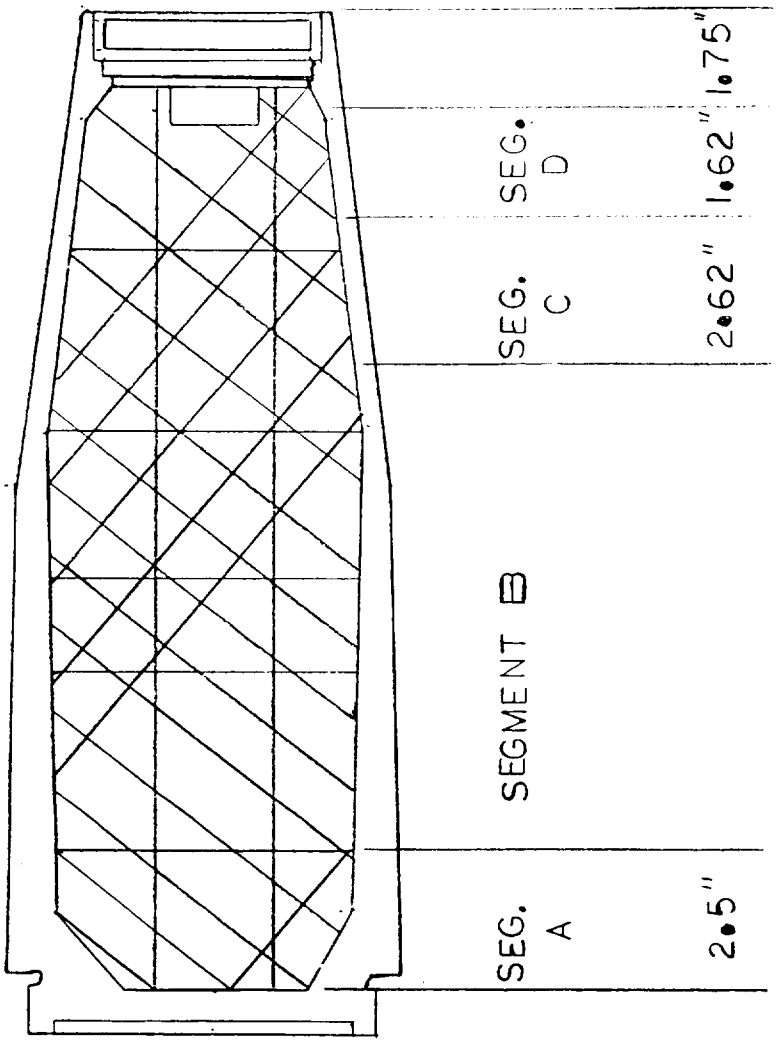


FIGURE 1. Segments

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3.3.2 Surface chipping. Chipping around the booster cavity shall not exceed one quarter of the sidewall area (1 sq. in. max.). Any warhead with excessive chipping shall be repaired by either probing or by drilling to a larger diameter and repouring the center of the cast (see 6.11).

3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. Not more than one transverse crack shall be permitted in Segment A and not more than a total of two transverse cracks shall be permitted in Segment A, B and C combined. Cracks shall be disregarded in Segment D.

3.3.5 Base separation. There shall be no base separation in excess of fifteen thousands of an inch (.015) between the base of the shell cavity and the cast when determined in accordance with 4.5.4.

3.3.6 Metal-parts defects. There shall be no crack, pipe, or thin wall section in the aft warhead.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge.

3.4 Loading of high explosive.

3.4.1 Thread surfaces. The surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the warhead.

3.4.2 Solid explosives. Solid explosive shall not be in the warhead before loading or added to the warhead during initial pouring or cooling.

3.4.3 Coating. Warheads having the interior wall imperfectly coated shall not be loaded. The interior surface of the warhead shall be dry when the high explosive is poured into the warhead (see 4.4.2.8).

3.4.4 Warhead interior cavity. The interior cavity of the warhead prior to loading shall be free from foreign matter.

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3.4.5 Unsolidified pool of paint. There shall be no unsolidified pool of paint in the bottom of the cavity prior to loading of the explosive.

3.4.6 Exterior surfaces. There shall be no explosive on the exterior surface of the projectile after loading.

3.4.7 Interior metal defects. The aft warhead cavity shall be free from embedded foreign matter, raised material or sharp edges.

3.5 Assembly torque. The motor body shall be assembled to The aft warhead and the forward warhead assembled to the aft warhead while the adhesive is still wet to meet the torque requirement specified on dwg. 9287994 per 4.4.2.14 and 4.4.2.15. The equipment used for applying torque to the motor body-warhead assembly and warhead/warhead assembly shall be calibrated at the start of each production shift as well as each time there is a change in equipment, settings or process that is likely to affect uniformity or performance as determined by the Government. The equipment shall be provided with a suitable indicator to aid in observance of proper torque application.

3.6 Proving ground test. The projectile shall comply with the following requirements when tested as specified in 4.4.3.4.

3.6.1 Functioning.

3.6.1.1 Rocket motor functioning. When the rocket cap is removed (rocket motor to function), the rocket shall fire in flight. The rocket motor shall ignite not earlier than four seconds after firing. Lots with early motor ignitions, which pass all other proving ground test requirements, shall be referred to the technical agency for disposition.

3.6.1.2 Horizontal range.

3.6.1.2.1 Range probable error. The pooled range probable error (RPE) of the ballistic samples fired as specified in 4.4.3.4.1 shall not exceed the appropriate value listed below:

Degrees-of-freedom	Maximum RPE (percent)
29	.364
28	.365
27	.366
26	.367
25	.370
9	.441
8	.454
7	.471
6	.495
5	.529

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3.6.1.2.2 Percent of mean horizontal range. No projectile shall have a horizontal range less than 97.5 percent of the mean horizontal range, unless the rocket motor fails.

3.6.1.3 Security. There shall be no premature bursts or indication of metal parts separation in the gun bore or in flight.

3.6.2 Safety. There shall be no premature bursts or indication of metal parts separation in the gun bore, or in flight, when fired at any of the conditions specified in 4.4.3.4. In addition, no projectile fired in the safety phase shall have a horizontal range less than 97.5 percent of the mean unless the rocket motor fails to ignite.

3.7 Inhibitor separation. The propellant grain shall contain no individual separation between the inhibitor and the grain or between the inhibitor layers (delamination) which is greater than 0.01 square inch. The total combined area of propellant separations and inhibitor delaminations shall be not greater than 0.02 square inch. There shall be no propellant inhibitor separation within one inch of any split surface.

3.8 Ring gaging. A ring gage manufactured to maximum bourrelet diameter shall pass over the warhead and rocket motor warhead joint after final touch-up painting.

3.9 Exudation. After post heat cycling, there shall be no exudation in the nose threads or on the exterior of the aft warhead.

3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. The requirements of workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.11.1 Cleaning. If required, the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.11.2 Protection of the rotating band. Care shall be exercised at all times so that the rotating band of each motor body will not be nicked or burred.

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3.11.3 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.11.4 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter, prior to inspection and shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requriements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspections shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Office for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Aft Warhead Loading Assy	9288022	10
Booster & Fwd Whd Assy	9288021	10
Projectile	9277994	20

4.3.2 Inspection to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile, 8 Inch, M650	1 OF 1	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Aft Warhead Loading Assy (Dwg. 928022)			
	Examination for Defects	10	3.2	4.4.2.8, 4.4.2.9
	Radiographic Inspection	10	3.3	4.4.3.1
	Booster and Fwd Whd Assy (Dwg. 9288021)			
	Examination for Defects	10	3.2	4.4.2.6
	Projectile (Dwg. 9277994)	(a)	3.2	4.4.2.1 thru 4.4.2.15
	Proving Ground Test	60	3.6	4.5.3
	Examination for Defects	62	3.2	4.4.2.15

NOTES: (a) Two of the projectiles submitted for first article shall be disassembled to allow checking of the examinations from 4.4.2.1 thru 4.4.2.15.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded projecitles shall contain:

a. Aft warhead metal parts of one lot interfix number from one manufacturer.

b. Forward warhead metal parts of one lot interfix number from one manufacturer.

c. Motor body metal parts of one lot interfix number from one manufacturer.

d. TNT from the same grade from one lot interfix number from one manufacturer.

e. Propellant grains from up to four consecutively produced lots of one lot interfix from one manufacturer.

f. Booster auxiliary assembly from one lot interfix number from one manufacturer.

g. Delay assemblies of one lot from one manufacturer.

h. Booster assemblies PA-524 from one lot interfix from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Delay Assembly and Motor Body Prior to Assembly		9287970/9288025
			NEXT HIGHER ASSEMBLY 9287992

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	O-ring missing	0.40%	3.2	Visual
102	Lubricant missing or insufficient from O-ring on outside of delay assembly	0.40%	3.2	Visual
103	Rocket motor body protective cap missing or incorrect	0.40%	3.2	Visual
104	Nozzle loose or missing	100%	3.2	Visual/Manual
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Delay Assembly After Assembly to Motor Body		9287992
			NEXT HIGHER ASSEMBLY 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ring seal missing or not pressed	0.40%	3.2	Visual
102	Grease on face of delay assembly	0.40%	3.2	Visual
103	Stripe showing orientation of delay	0.40%	3.2	Visual
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Rocket Cap Assembly, Prior to Assembly to Motor Body	1 OF 1	9288040
			NEXT HIGHER ASSEMBLY 9287992

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Rocket Cap seal ring missing	100%	3.2	Visual
102	Rocket Cap O-ring missing	100%	3.2	Visual
103	Lubricant on face of Rocket Cap Assembly	0.40%	3.2	Visual
104	Lubricant not applied to O-ring	100%	3.2	Visual
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9287992
4.4.2.4	Motor Body and Delay Assembly After Assembly of Rocket Cap		NEXT HIGHER ASSEMBLY 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Rocket Cap not assembled with proper torque	100%	3.2	Gage
Minor	None defined			

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Rocket Motor Prior to Loading Propellant Grain	1 OF 1	9287994
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Nozzle insert missing, or damaged after assembly of delay	100%	3.2	Visual
Major				
101	Foreign matter in cavity	0.40%	3.2.1	Visual
102	Grain support missing or loose	0.40%	3.2	Visual
103	Aft cushion improper, missing or not bonded to grain support	0.40%	3.2	Visual
104	Propellant grain spacers not properly assembled or missing	0.40%	3.2	Visual
105	Inhibitor separation	0.40%	3.7	Visual
106	Presence of any chips, cracks, gouges, dirt, grease or other foreign material on surface of the propellant grain	0.40%	3.2.3	Visual
Minor				
201	Evidence of poor workmanship	1.0%	3.12 excluding 3.12.4	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Booster Auxiliary PA-500 & Forward Warhead Section Assembly	1 OF 1	9288021
			NEXT HIGHER ASSEMBLY: 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Booster missing, damaged, improperly assembled or not held securely in place	0.40%	3.2	Visual/Manual
102	Minimum depth to top of booster	100%	3.2	Gage
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

NOTES:

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Aft Warhead Prior to Load With Explosive	1 OF 1	9288022
			NEXT HIGHER ASSEMBLY: 928/993

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Foreign matter in charge cavity	100%	3.4.4	Visual
2	Unsolidified pool of paint or blisters in bottom of cavity	100%	3.4.5	Visual/Manual
3	Interior metal defects	100%	3.4.7	Visual
Major				
101	Interior protective coating with bare spots	0.40%	3.4.3	Visual
102	Forward protective cap missing or improper	0.40%	3.2	Visual
103	Aft protective cap missing or improper	0.40%	3.2	Visual
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.8	Aft Warhead Section: Loading Assembly		9288022
			NEXT HIGHER ASSEMBLY 9287993

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Cavities and cracks in high explosive charge	100%	3.3	4.4.3.1/X-ray
2	Base separation	(a)	3.3.5	4.4.3.1.1
3	Presence of explosive on exterior of shell body including threads, and mating surface to forward warhead and motor body	100%	3.4.6	Visual
4	Presence of crack, pipe or thin wall section in the aft warhead	100%	3.3.6	4.4.3.1.2
5	Foreign material in explosive charge	100%	3.3.7	4.4.3.1.3
Major				
101	Explosive charge not properly faced off	0.40%	3.2	Gage
102	Foreign matter in warhead nose	0.40%	3.2	Visual
103	Booster missing or not securely bonded	0.40%	3.2	Visual/Manual
104	Foreign matter on shoulder undercut or threads	100%	3.4.1	Visual
105	Booster cavity depth incorrect	0.40%	3.2	Gage
106	Booster cavity diameter incorrect	0.40%	3.2	Gage

NOTES: (a) See paragraph 4.4.3.1.1.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Aft Warhead Section: Loading Assembly	2 OF 2	9288022
			NEXT HIGHER ASSEMBLY 9287993

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Excess chipping on exposed explosive surface in nose of warhead	100%	3.3.2	Visual
202	Protective coating with total damaged areas in excess of 1/4 square inch	0.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable including incorrect color code	0.65%	3.2	Visual
204	Evidence of poor workmanship	1.0%	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Aft Warhead Section: Loading Assembly Prior to Assembly of Pad	1 OF 1	9287993
			NEXT HIGHER ASSEMBLY: 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical 1	Filler or other foreign matter on shoulder, undercut or threads in warhead nose	100%	3.2	Visual
Major 101	Booster missing	100%	3.2	Visual
Minor 201	Evidence of poor workmanship	1.0%	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.10	Aft Warhead Section: Loading Assembly Prior to Assembly of Disk & Disc Assembly		9287993
			NEXT HIGHER ASSEMBLY 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pad missing	0.40%	3.2	Visual
102	Adhesive missing or insufficient on disk (disc) prior to assembly of disc (disk)	0.40%	3.2	Visual
103	Disc not assembled concentric to disk (allowing improper seating in aft warhead)	100%	3.2	Visual/Templet
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Aft Warhead Section: Loading Assembly Prior to Assembly of Booster Auxiliary PA-500 and Forward Whd. Section Assembly	1 OF 1	9287993
			NEXT HIGHER ASSEMBLY: 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Filler on exterior of warhead	100%	3.2	Visual
Major				
101	Aft warhead or forward warhead threads not free from grease or foreign matter prior to application of sealant	0.40%	3.2	Visual
102	Disk, disc or lock washer missing or improperly assembled	0.40%	3.2	Visual
103	Sealant missing or improperly applied to threads	0.40%	3.2	Visual
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Warhead, 8 Inch, HE: M650	1 OF 1	9287993
			NEXT HIGHER ASSEMBLY: 9287994

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Gap at aft warhead - forward warhead interface (.002 Feeler)	100%	3.2	Gage
102	Lifting plug gasket missing	0.40%	3.2	Visual
103	Lifting plug not fully seated	0.40%	3.2	Visual
104	Aft warhead - forward warhead assembly torque	100%	3.5	4.4.3.5/Test
105	Reapplication of assembly torque to aft warhead - forward warhead	0.40%	3.5	4.4.3.5/Test
106	Lifting plug disassembly torque	0.40%	3.2	4.4.3.6/Test
107	Gap at aft warhead - forward warhead interface (.005 Feeler)	100%	3.2	Gage
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Rocket Motor and Warhead Prior to Assembly	OF 1	9287994
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Threads and bearing surfaces free from grease, explosive, or foreign material, prior to application of sealant	100%	3.2	Visual
Major				
101	O-ring missing from warhead	100%	3.2	Visual
102	Warhead insulation missing	100%	3.2	Visual
103	Sealing compound missing or improperly applied	100%	3.2	Visual
104	Inhibited end of grain not facing aft	100%	3.2	Visual
105	Forward grain cushion improper, missing or improperly lubricated, prior to application of sealant	0.40%	3.2	Visual
106	Lubricant missing from O-ring	0.40%	3.2	Visual
107	Aft protective cap missing or improper on aft warhead	0.40%	3.2	Visual
Minor				
201	Evidence of poor workmanship	1.0%	3.12	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Projectile, RA, 8 Inch HE: M650	1 OF 2	9287994
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Obturator missing or damaged	100%	3.2	Visual
2	Projectile zone weight incorrect	100%	3.2	Balance
3	Zone stencilling or punch marks missing, inadequate or improper	100%	3.2	Visual
4	Gap between warhead and motor body assembly	100%	3.2	Gage
Major				
101	Missing or improperly assembled components	100%	3.2	4.4.3.3/X-ray
102	Step up condition from warhead to rocket motor over maximum	100%	3.2	Gage
103	Rocket motor-warhead assembly torque	100%	3.2	4.4.3.2/Test
104	Reapplication of assembly torque to rocket motor - warhead joint	-	3.2	4.4.3.2/Test
105	Rotating band nicked or damaged	0.40%	3.2.4	Visual
106	Projectile fails to pass through ring gage	0.40%	3.2	Gage
107	Weight zone punch mark not discernible after painting	0.40%	3.2	Touch
108	Torque test failure of rocket-off cap to be checked at the last station (prior to palletizing)	0.40%	3.2.2	Test/4.4.3.9

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Projectile, RA, 8 Inch HE: M650	2 OF 2	9287994
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Painted surface with total exposed or damaged area in excess of 1/4 sq. in.	0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable including incorrect color code	0.65%	3.2	Visual
203	Grommet missing, damaged or improperly assembled	0.65%	3.2	Visual
204	Lifting plug improperly lubricated	0.65%	3.2	Visual
205	Evidence of poor workmanship	1.0%	3.12	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Pallet	1 OF 1	9280132
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Cover support missing (6 places)	100%	3.2	Visual
Major				
101	Pallet damaged or improperly assembled	0.40%	3.2	Visual
102	National stock number or DOD code missing or unidentifiable	0.40%	3.2	Visual
103	Strapping or staples missing or loose	0.40%	3.2	Visual
104	Projectiles of different weight zones on the same pallet	0.40%	3.2	Visual
Minor				
201	Number of projectile assemblies incorrect	0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable	0.65%	3.2	Visual
203	Strapping or staples improperly engaged	0.65%	3.2	Visual
204	Evidence of poor workmanship	1.0%	3.12	Visual

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4.4.3 Testing.

4.4.3.1 Cavitation (see 3.3 except 3.3.2). Cavitation defects are classified as follows:

Critical:

1. Cavitation, porosity, piping or annular rings in Segment A in excess of that permitted in Table I..... X-ray

2. Cavitation, porosity, piping or annular rings in Segment B or C in excess of that permitted in Table I..... X-ray

3. Cracks in excess of that permitted in 3.3 and 3.3.4..... X-ray

4. Base separation (see 4.4.3.1.1)..... X-ray

5. Cavitation in Segment D in excess of that permitted in Table I..... X-ray

This test shall be performed 100 percent on all warhead section loading assemblies. Each assembly shall be subject to X-ray examination. The test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4. Any assembly which has any of the above critical defects shall be classed defective and removed from the lot.

Warheads rejected by radiographic examination may be drilled out to one half the depth of the "B" segment in Figure 1 and repoured, but not more than one drilling may be performed per warhead.

4.4.3.1.1 Base separation (see 3.3.5). Warheads showing the largest base separation as determined by X-ray examination prior to post cycle heat treatment shall be selected for sectioning at the following rate. Select warheads at the rate of one (1) out of seven hundred (700) warheads to be included in the post cycle heat treatment lot with a minimum of two (2) samples per lot. These warheads are to be included in the post cycle heat treatment and sectioned for determination of base separation thereafter. If insufficient samples are available with base separation, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. The samples selected for sectioning shall be placed in the coolest area of the post cycle igloo.

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Any warhead after sectioning which has base separation in excess of that permitted by the applicable requirement, shall be classed defective, and the entire post cycle heat treatment lot rejected. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.4. Additionally, no warhead lot shall be post cyclic heat treated more than three times.

4.4.3.1.2 Metal parts defects. All radiographs of warhead section loading assemblies shall be examined for evidence of crack, pipe and thin wall sections in the aft warhead. The presence of any such indications shall cause that assembly to be classed defective and removed from the lot (see Section 6.14).

4.4.3.1.3 Foreign material. All radiographs of warhead section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any foreign material shall cause that assembly to be classed defective and removed from the lot.

4.4.3.2 Motor body - warhead assembly torque (see 3.5) - Major Defect. The torque applied to the motor body-warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hours production for a check torque application. If any projectile fails to comply with the torque requriements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check shall be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected.

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4.4.3.3 X-ray examination for missing or improperly assembled components (see dwg. 9287994). This test shall be performed 100 percent. Each assembly shall be subjected to an X-ray examination in a plane that will clearly define all of the motor body components, including both channels of the delay assembly. Necessary controls will be implemented to assure that the assembly will be oriented such that both channels are visible when x-rayed. The test shall be performed as specified in 4.5.1.2 with equipment in accordance with 4.4.4. Any assembly which shows evidence of missing or improperly assembled components, as listed below, shall be classed as defective and removed from the lot.

a. Delay housing assembly not properly seated in motor body.

b. Shear ring seal not fully expanded in delay and motor body grooves. The groove in the motor body shall be completely filled with lead. There shall be no gap between the shear ring seal and the shoulder of the expansion cone.

c. Boron potassium nitrate charge missing.

d. Voids or missing increments in the delay column.

e. Evidence of foreign material present.

f. Any other deviation from the normal assembly condition.

4.4.3.3.1 Radiographic film. Only a film type and vendor as listed herein has been tested and approved by AMCCOM for use in radiographing this projectile (see 6.9.1). A substitute item shall not be used before prior testing or approval.

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4.4.3.4 Proving ground (see 3.6). These tests shall be performed as specified in 4.5.3. Defects shall be classified as follows:

Categories	Defects
Critical:	
1	Premature burst
Special:	
a	Range less than 97.5 percent of mean horizontal range unless rocket motor fails to ignite (see 6.16).
b	Metal parts separation (see 6.17).
Major:	
101	Range probable error exceeds allowable limit (see 3.6.1.2.1).
102	Rocket motor fails to ignite.
Minor:	None defined

4.4.3.4.1 Functioning (see 3.6.1). The functioning test shall be conducted in accordance with 4.5.3.1.

4.4.3.4.1.1 Sample size. For first article or beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria specified herein, and in 4.4.3.4.2.1, 30 sample projectiles shall be selected at random from each lot for this test and fired with the rocket cap removed (rocket motor to function). If any critical or major defect occurs, the lot shall be rejected.

4.4.3.4.1.2 Successive lots. After three consecutive lots have complied with 4.4.3.4.1.1, 10 sample projectiles shall be selected at random from each successive lot for this test, and fired with the rocket cap removed (rocket motor to function). If any critical or major defect occurs, the lot shall be rejected. In the event that two successive lots fail to meet any or all of the requirements of this paragraph or 4.4.3.4.2.2, the provisions of 4.4.3.4.1.1 and 4.4.3.4.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.4.2 Safety (see 3.6.2). The safety tests shall be conducted in accordance with 4.5.3.2.

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4.4.3.4.2.1 Sample size. Beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria contained herein and in 4.4.3.4.1.1, 30 sample projectiles shall be selected at random from each lot for these tests and fired with the rocket cap removed (rocket motor to function). If any critical defect occurs, the lot shall be rejected.

4.4.3.4.2.2 Successive lots. After three consecutive lots have complied with 4.4.3.4.2.1, 14 sample projectiles shall be selected at random for these tests, and fired with the rocket cap removed (rocket motor to function). If any critical defect occurs, the lot shall be rejected. In the event that two successive lots fail to meet the requirements of this paragraph or 4.4.3.4.1.2, provisions of 4.4.3.4.1.1 and 4.4.3.4.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.4.2.3 Temperature test failures. Failure of a rocket motor to ignite during +145°F safety tests shall be considered a Major defect, and the lot shall be rejected. If a motor ignition failure occurs during -50°F safety tests, the lot shall be referred to the procuring agency for disposition (see 6.14).

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4.4.3.4.3 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.3.5 Forward warhead - Aft warhead assembly torque (see 3.5). The torque applied to the forward warhead - aft warhead joint shall be witnessed for each projectile during assembly to assure compliance with the applicable requirement. Any projectile improperly assembled shall be classed defective and removed from the lot. In addition, a minimum of one sample projectile shall be randomly selected from each hour's production for a check torque application. If any projectile fails to comply with the torque requirements, as evidenced by relative motion at the joint, when tested in accordance with 4.5.2 then all projectiles assembled since the last acceptable sample shall be rejected.

As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four-hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine display shall be checked during operation on a subsequent assembly. If the indicated torque is in or below the lower 20% of the acceptance range, a verification check shall be made. If the machine fails the verification, all projectiles assembled since the last acceptable verification shall be rejected.

4.4.3.6 Lifting plug disassembly torque. A thin line shall be drawn across the joint in question to aid in observing movement. If any projectile fails to comply with the torque requirement per 4.5.2, the lot shall be rejected.

a. Lifting plug (loosening at min disassembly torque)

b. Lifting plug (failure to loosen at max disassembly torque)

4.4.3.7 Exudation - Minor Defect. Six aft warheads from the hottest area of the post cycle oven shall be used for the sample. any aft warhead with exudation shall be cause for rejection of the entire post cycle treatment lot.

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4.4.3.8 Grain inspection. If the grains are received in packing sleeves which are used for assembling the grains directly into the motor bodies, then visual inspection of the grains shall be conducted without moving the grains from the sleeves (i.e., only exposed surfaces to be inspected).

4.4.3.9 Rocket cap. Projectiles received at this inspection station without a rocket cap shall not have a cap assembled, but shall be rejected and removed from the lot. Delay assemblies must be extracted and replaced.

4.4.4 Inspection equipment. Requirements for inspection equipment to perform examinations and tests are specified in the item detail specification and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection equipment designs.

4.5 Methods of inspection.

4.5.1 X-ray examination.

4.5.1.1 Determination of aft warhead defects. The warheads shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the warhead and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453 (see 4.5.1.3). In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.4 and 4.5.1.5. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as film or processing related or as acceptable metal parts irregularities.

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In the event an original radiograph leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location, or classification of a defect, the projectile shall be rotated about its longitudinal axis from the original exposure, and additional radiographs taken for clarification. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where radiograph reveals superimposed cavities, each of which individually complies with the requirements. All additional radiographs of a shell for defect conditions will be identified on the film as retakes and stored; traceable to the original.

4.5.1.2 Determination of missing or improperly assembled motor body components. In addition to the warhead examination required by 4.5.1.1, the motor body section of the assembled projectile shall be subject to radiographic examination (see 4.4.3.3). In all exposures the radiographic beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination shall be made to determine compliance with applicable requirements.

The aiming point of the beam shall be determined based on standard radiographs (see 4.5.1.6) which give most accurate results, and once the particular beam location is established, it shall not be changed unless new radiographic film standards are established and reviewed by the technical agency (see 4.5.1.6).

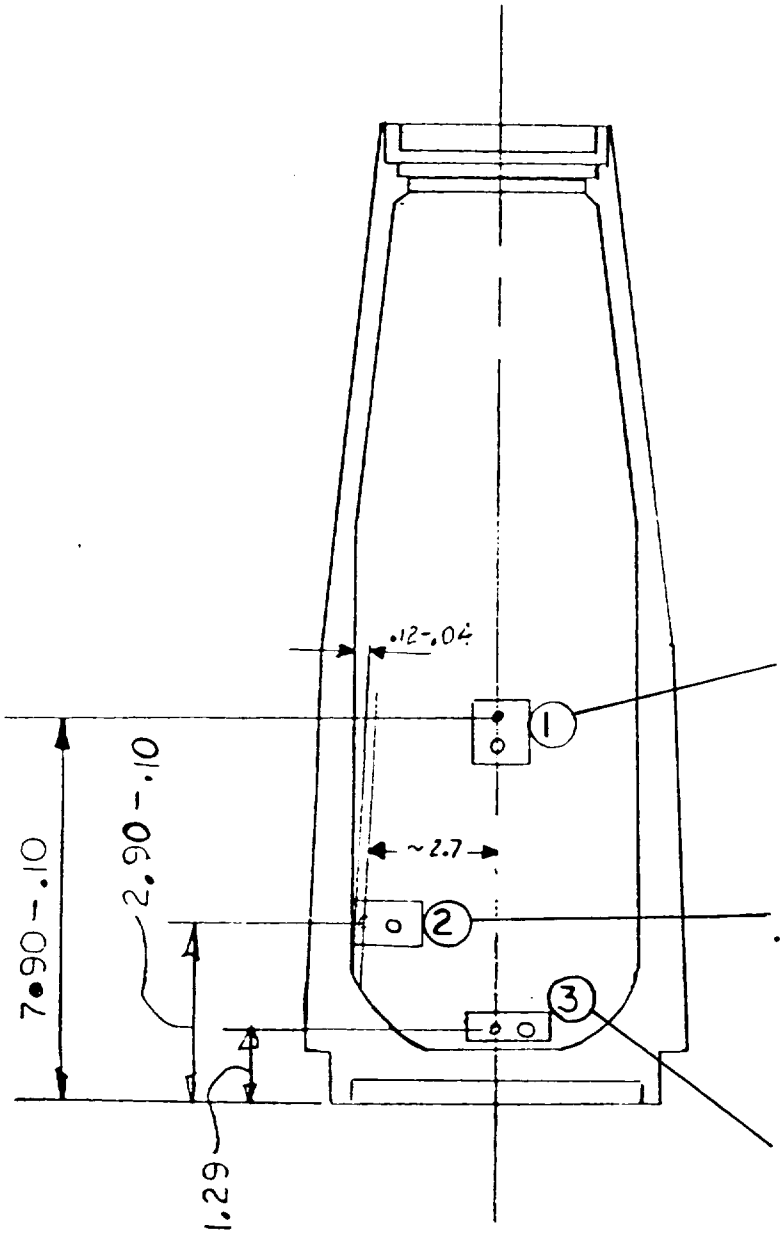
4.5.1.3 Radiographic equipment, operations and procedures. Radiographic equipment, operations and procedures shall be qualified in accordance with Standard MIL-STD-453.

4.5.1.4 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of a warhead so that the 2T holes are at the points shown in Figure 2. Three penetrameters for each warhead shall be used. When radiographing a number of similar warheads, the frequency of the sensitivity check shall be not less than one in each fifty radiographs, and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring.

The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-453.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameters see MIL-STD-453.

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Penetrameters Consturcted Per MIL-STD-453. Dimensions shown are for loaciton of center of 2T holes.

Ø.80" single steel wall plus 6.40" explosive. Penetrameter thickness=.060" 2T hole diameter - Ø.120".

2.10" single steel wall plus 1.76" explosive at .120" from tangent point. Penetrameter thickness-Ø.090"

2.5" single steel wall plus 3.0" explosive. Penetrameter thickness=Ø.110" 2T hole diameter = Ø.220".

NOTE: Penetrameters number 2 and 3 shall be positioned such that the images of the 2T holes appear tangent to the inner steel wall of the warhead.

FIGURE 2. Radiographic penetrameter requirements.

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4.5.1.5 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. (Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units.) Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of not more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photograhic step tablet calibrated by National Bureau of Standards or approved by ARDEC. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.6 Review of radiographic sensitivity and technique (see 6.9). The sensitivity of the radiographic techniques shall be reviewed by submitting one x-ray of an aft warhead with penetrameters per 4.5.1.4 and a copy of radiographic procedures prior to the start of initial production. In addition, an x-ray of the motor body section of an assembled projectile shall be randomly selected and submitted for review of the sensitivity of the inspection for proper assembly. These x-rays shall be taken and processed using the same techniques as will be used in production. In addition, one x-ray of the warhead standards and an x-ray of the motor body section shall be submitted for review once a month for the first three months of each contract and every three months thereafter.

4.5.1.7 Operator qualifications. For operation of radiographic equipment, personnel shall be qualified and certified in accordance wtih MIL-STD-410, Level I. For setup and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

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4.5.2 Torque tests. These tests shall be performed on the sample selected in accordance with 4.4.2.14 and 4.4.2.15. A thin line shall be drawn across the joint in question to aid in observing movement. The minimum assembly torque specified on the drawing for the rocket cap and the motor body and warhead sections shall be applied in the assembly direction. The minimum and maximum disassembly torques for the lifting plug shall be applied in the disassembly direction.

4.5.3 Proving ground test. This test shall be performed at a Government proving ground in accordance with the applicable Acceptance Test procedure. the samples selected in accordance with 4.4.3.4 shall be fired in a M201 series cannon with muzzle break. The projectiles shall be fuzed with a M557 series PD fuze. Adequate photographic and instrumental coverage shall be provided to assure compliance with all applicable requirements. If projectiles are received with "ballistic sample numbers" from the contractor then the projectiles shall be randomized with regard to test phase and order of firing. The following information pertaining to this test shall be obtained and reported for all phases as specified below.

a. Weapon and component data. All weapon and ammunition components used in support of this test shall be identified by model and lot or serial numbers. This information shall be reported on the firing record.

b. Observable attribute data shall be reported as follows:

1. Premature functioning.
2. Metal parts separation.
3. Rocket motor malfunction.
4. Impact functioning (information only).

c. Variable round-by-round data. Data specified in the Firing Record Information Sheets, see Figures 3 and 4 for sample forms, shall be obtained for each ballistic sample.

d. Meteorological (Met) data. Met data shall be obtained to an altitude of 14,000 meters at the start of testing (± 15 min.) and at least every two (2) hours throughout the test.

e. Radar data. Doppler radar (or a suitable alternative with Government approval) shall be obtained for each round fired until radar tracking cannot be maintained. Raw radar tapes shall be maintained a minimum of six months after release of the firing record.

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f. Photographic coverage. Photographic coverage shall be adequate to assure compliance with the applicable requirements. Each projectile ballistically tested shall have four lines painted across the aft warhead/motor body joint and across the forward warhead/aft warhead joint (see 6.10). Photographic coverage shall be provided to determine relative movement between joints. Additionally, a second camera shall be positioned to ascertain ignition (burning) of the delay assembly columns.

4.5.3.1 Functioning. all test and information control projectiles shall be conditioned and fired at 70 degrees Fahrenheit (F) using a Charge 9, M188 series propelling charge conditioned at 70°F. In order to minimize the effects of met conditions, the test shall be fired in either one (for normal inspection) or three (for tightened inspection) series of 10 rounds each. The first subgroup for normal inspection or two subgroups for tightened inspection shall be fired at a quadrant elevation of 30 degrees and the last subgroup shall be fired at a quadrant elevation of 52 degrees. Each 10 round subgroup shall be immediately preceeded by two (2) informational control rounds and, in addition, the final subgroup shall be followed by two (2) information control samples. Time lapses of more than 15 minutes between consecutive rounds shall not be permitted and it is desirable that each 12 round series (2 informational control samples plus 10 test samples) be completed within one (1) hour. Range probable errors shall be calculated for each 10 round subgroup and then a pooled range probable error shall be computed for determination of compliance with 3.6.1.2.1.

$$\text{Subgroup RPE (percent)} = RPEj \sqrt{\frac{\sum_{i=1} (X_i - X)^2}{n-1} \times \frac{(100 \times .6745)}{\text{Ave. Range}(j)}}$$

$$\text{Pooled RPE (percent)} - RPEp = \sqrt{\frac{(RPE_1)^2 (n_1 - 1) + (RPE_2)^2 (n_2 - 1) + (RPE_3)^2 (n_3 - 1)}{DF}}$$

Where X = Individual horizontal ranges within each subgroup.

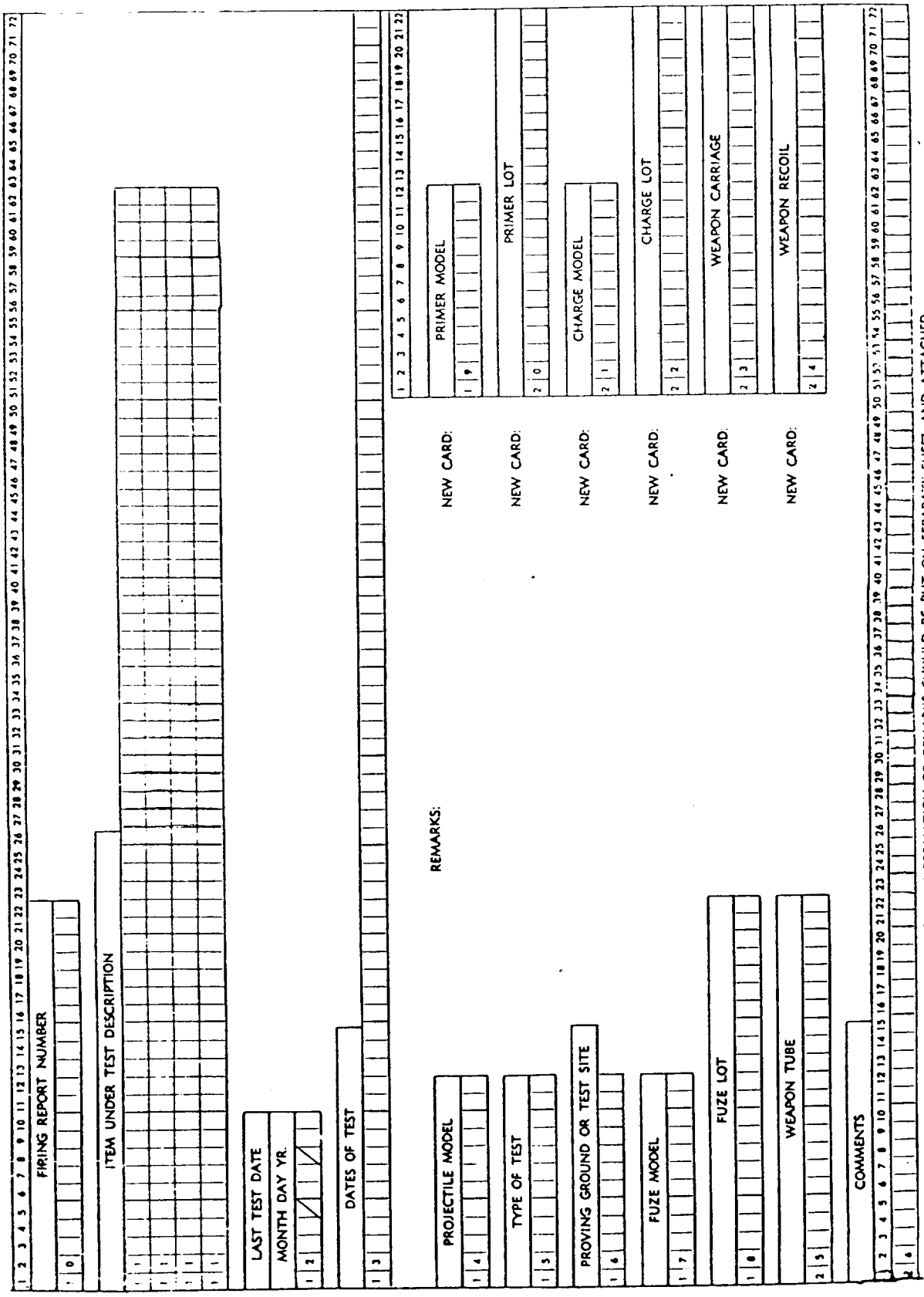
Where DF = Degrees of freedom n(1) + n(2) ...n(j) - number of suggroups.

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FIGURE 3

M650 FIRING RECORD INFORMATION

OVERALL DATA



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FIGURE 4

Sheet 2

M650 FIRING RECORD INFORMATION FOR FIRING RECORD NUMBER

FILL IN

NOTE: A NEW SHEET 2 MUST BE STARTED EACH TIME THERE IS A CHANGE IN QUADRANT ELEVATION, PROJECTILE TEMPERATURE, PROPELLANT TEMPERATURE, DATE FIRED OR PROJECTILE LOT.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

Enter different letter code for each Firing Data Form (A, B, etc.) for same firing record

0 1

PROJECTILE LOT

3 9

LOT (LEAVE BLANK)

0 2

QUADRANT ELEVATION (DEGREES)

4 1

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

DATE OF FIRING: MONTH - DAY - YEAR

4 0 / /

PROJECTILE TEMPERATURE DEGREES F.

4 3 .

PROPELLANT TEMPERATURE DEGREES F.

4 4 .

TYPE CARDS ONLY WHERE THERE ARE ENTRIES

	TUBE RD. NO.	SAM. NO.	PROJECTILE WEIGHT	TIME FIRED	MUZZLE VEL. (MPS)	AVG. CH. PR. (PSI X 100)	RKT. IGN. (SECS.)	HORIZONTAL RANGE (M)	DEFLECTION (M)
1 2	3 4 5 6	7 8	9 10 11 12 13	14 15 16 17	18 19 20 21 22 23	24 25 26 27 28	29 30 31 32 33	34 35 36 37 38	39 40 41 42 43
0 0			.		.		.		
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COMMENTS AND REMARKS:

DRDAR-QA FORM 530-1, 1 MAR 80 REPLACES EDITION OF 1 JUN 78 WHICH MAY BE USED UNTIL EXHAUSTED

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4.5.3.2 Safety. The projectiles for this test shall be divided into two equal groups. Group 1 shall be conditioned and fired at -50°F for 24 hours at Charge 9, M188 series propelling charge conditioned at -50°F for 24 hours. Group 2 shall be conditioned and fired at 145°F for 24 hours at Charge 9, M188 series propelling charge conditioned at 145°F for 24 hours. The cold phase (Group 1) shall be fired at a quadrant elevation of 52 degrees and the hot phase (Group 2) may be fired at a quadrant elevation to 53 degrees.

4.5.4 Determination of base separation. After post cycle heat treatment, the samples from each post cycle heat treatment lot shall be cooled to ambient and sectioned in a manner to expose the interface between the cast and the interior base of the aft warhead at the approximate center of the base. The aft warhead shall be positioned nose down or the cast shall be moved as far towards the nose with finger pressure as possible before measuring for base separation between the shell cavity and cast.

4.5.5 Ring gage. After touch-up and joint painting, projectiles shall be passing through a ring gage as specified on Dwg. 9287994. The ring gage is to pass over the bourrelet and rocket motor warhead joint.

4.5.6 Exudation. The exterior of the aft warhead shall be visually inspected for exudation. If no exudate is found, remove the forward protective plug and inspect for exudation visually.

5. PACKAGING

5.1 Packing - Level A. The projectile shall be packed in accordance with Dwg. 9280132.

5.2 Marking. Marking shall be in accordance with Dwg. 9280132, MIL-STD-129, and Code of Federal Regulations - Title 49, Transportation parts 100-199.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components and assemblies covered by this speicification are intended for use on the M650 R.A. Projectile.

6.2 Acquisition requirements. See MIL-A-48078 (AR).

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6.3 Submission of inspection procedures and equipment design for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Photographic densities. "American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, has been satisfactory as a source of information on film densities.

6.5 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and coolinng operations and scrap formed during the drilling operation.

6.6 Inspection lot size. It is noted that the size of inspection lots of components assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

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6.7 Proving ground test summary.

Test	Sample Size (see 4.4.3.4)		Requirement
Functioning	*30	10	a. Premature burst or metal parts separation (see 3.6.1.3). b. Individual range (see 3.6.1.2.2). c. Rocket Motor function (see 3.6.1.1). d. Range standard deviation (see 3.6.1.2.1).

*Applicable only for first three lots or tightened inspection.

Test	Sample Size (see 4.4.3.4)		Requirement
Safety (sample divided into 2 groups and fired hot and cold as per 4.3.2.)	*30	14	a. Premature burst or metal parts separation (see 3.6.2). b. Individual range (see 3.6.2). c. Rocket motor funciton (see 3.6.2).
TOTAL SAMPLE PROJECTILES REQUIRED			
	*60	24	

*Applicable only for first three lots or tightened inspection. Informational-control projectiles are not included.

6.8 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for loaded items provided by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commander, AMCCOM, Picatinny Arsenal, New Jersey 07806-5000, ATTN: AMSMC-QAR-Q(D) and ARDEC, Picatinny Arsenal, New Jersey 07806-5000, ATTN: SMCAR-FSA-PS.

6.9 Radiographic procedures. Radiographic procedures, standards and sample radiographs shall be submitted to the Commander, ARDEC, Picatinny Arsenal, New Jersey 07806-5000, ATTN: AMSMC-QAR-Q(D) for review and approval.

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6.9.1 Radiographic film.

a. Eastman Kodak Company
1187 Ridge Road, West
rochester, New York 14650

Film Types: AA, T, or M

b. E. I. DuPont de Nemours and Company
Photo Products Departments
Wilmington, Delaware 19898

Film Type: NDT-55, NDT-65 or NDT-70

c. GAF Corporation
140 West 51st Street
New York City, New York 10020

Film Type: 800 or 400

6.10 Ballistic test projectiles. All projectiles tested by a Government Proving Ground for Ballistic Acceptance Test shall have four (4) equally spaced lines painted, at the LAP manufacturing facility, across the aft Warhead/Motor Body joint and across the forward warhead/aft warhead joint using Stencil Ink, White No. 37875 per Spec TT-I-1975. These lines shall be approximately one half (1/2) inch wide by six (6) inches long and positioned with three (3) inches on the Warhead and three (3) inches on the Motor Body and on both sides of the forward warhead/aft warhead joint.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frandford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.12 Rocket motor ignition failure at -50°F. Degradation of rocket motor ignition reliability is expected at temperatures below -25°F. When an ignition failure occurs at -50°F during acceptance testing, the procuring agency must evaluate the following prior to determining lot disposition:

a. The delay assembly shall have functioned properly as evidenced by review of radar data.

b. The failure shall not be related to a change in components or processes.

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c. The effect on compatibility with the M753 NUC round must be considered. This may require a retest at -30 F.

d. The rate of failure should not be significantly different from prior experience.

6.13 Notification of defects. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of radiographs. The rejected projectile will be held pending disposition by the procuring activity.

6.14 Short range. Projectiles failing to achieve required range with proper motor function shall be considered critical defects. Projectiles failing to achieve required range due to improper motor ignition delay or motor burn shall be considered as major defects.

6.15 Metal parts separation. Metal parts separations which result in warhead impact short of 6000 meter safety zone for rocket-on fire, or which are not caused by rocket motor action, shall be considered critical defects. Metal parts separations caused by rocket motor action which result in warhead impact within the safety zone shall be considered major defects.

6.16 Subject term (key word) listing.

Assembly torque
Cavitation in HE charge
Delay
Explosive scrap
Motor body
Propellant grain
Rotating band

6.17 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-A960)

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